CANDIDATE NAME: ________________________________ INDEX NUMBER _________

CG _________

SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

H2 BIOLOGY **9744/02**
Paper 2 Structured Questions

Thursday
13 September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CG in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
Paper 1 (MCQ)	/30
Paper 2	
1	/18
2	/17
3	/11
4	/13
5	/12
6	/9
7	/8
8	/12
P2 Total	/100
Paper 3	/75
Paper 4	/55
TOTAL (100%)	

This question paper consists of **28** printed pages including this cover page.

1. Fig. 1.1 is an electron micrograph of part of an animal cell obtained from a rodent.

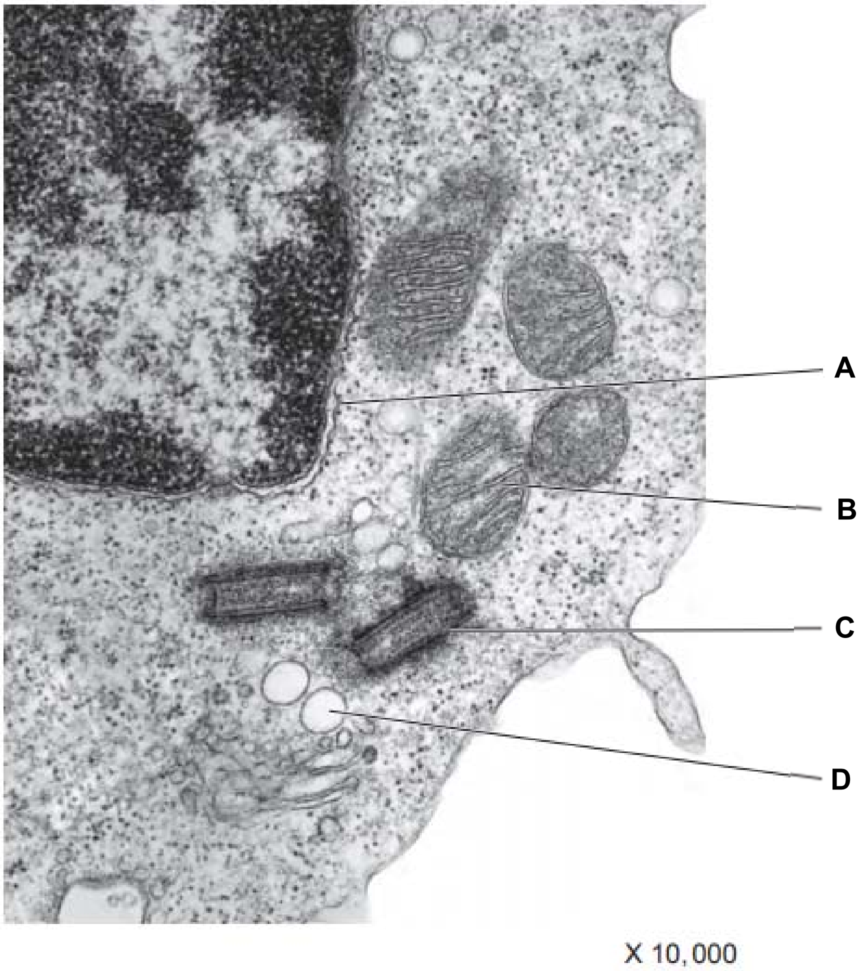

Fig. 1.1

(a) Name the structures labelled **A** to **C**.

A: ……………………………………………………………………………………………………..

B: ……………………………………………………………………………………………………..

C: …………………………………………………………………………………………………… [3]

(b) Describe the role of structure **C** in animal cells.

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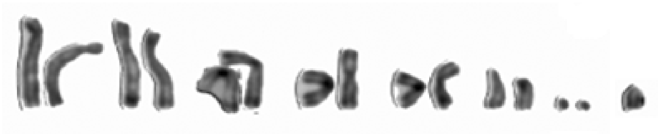
(c) Complete karyotypes of two rodents of the same species were isolated as shown in Fig. 1.2. In this species of rodent, males are the heterogametic sex, where they have two different sex chromosomes.

Normal Rodent



Mutant Rodent

Fig. 1.2

(i) State the diploid number of chromosomes in a normal rodent.

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(ii) Explain how the mutant rodent karyotype was formed.

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(iii) Suggest why the mutant rodent is still able to survive despite the mutation.

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Fig. 1.3 is an electron micrograph of a cancer cell in the midst of mitosis.

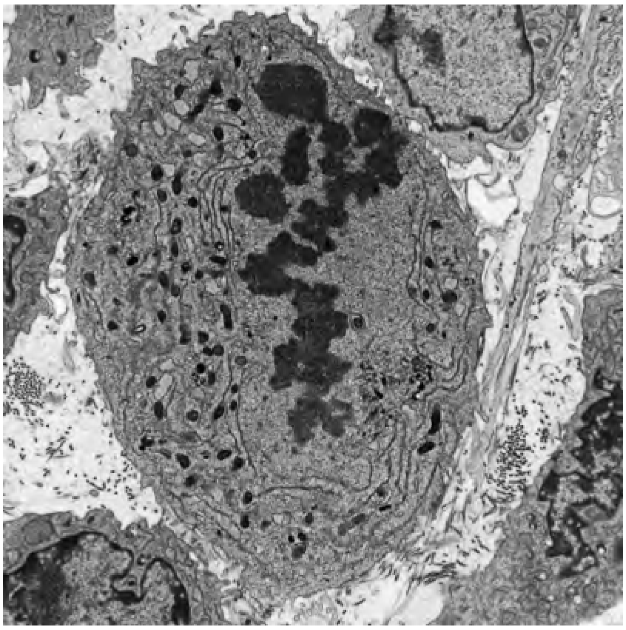

Fig. 1.3

(d) Identify the stage of mitosis, giving a reason for your answer.

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People who have smoked cigarettes for many years are at risk of developing lung cancer.

(e) Explain why smoking increases the risk of cancer.

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Fig. 1.4 shows the change in the percentage of smokers in the male population of the UK between 1950 and 2005. Fig. 1.5 shows the change in mortality rate in the UK in men aged 75 to 84 between 1950 and 2005.

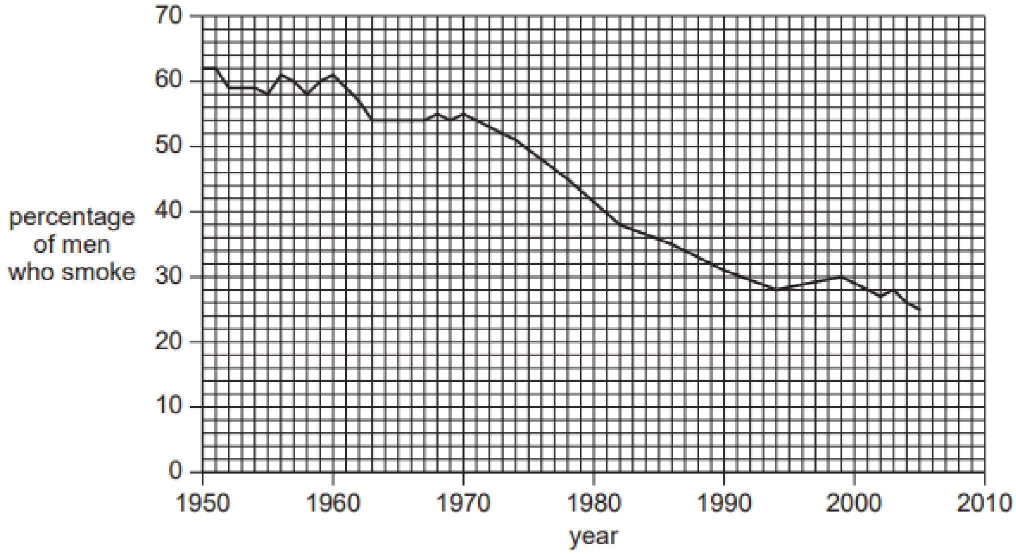

Fig. 1.4

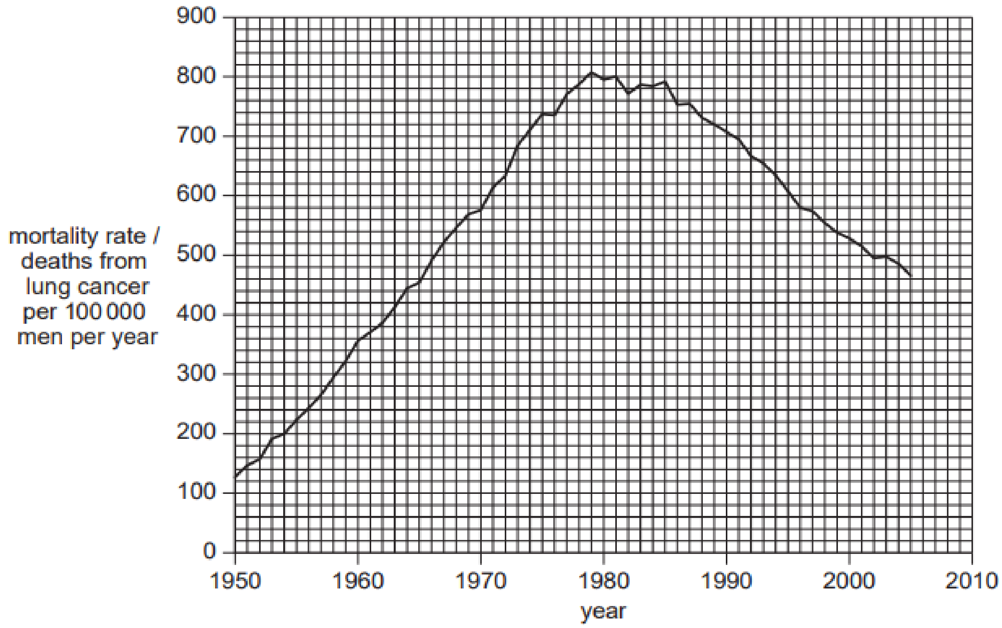

Fig. 1.5

(f) With reference to Figs. 1.4 and 1.5, discuss the observations made between 1950 and 2005.

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[Total: 18]

2. **(a)** Contrast between the structures of DNA and tRNA.

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.... [3]

(b) Table 2.1 shows two mRNA triplets. Fill in the complementary tRNA triplets in the spaces provided.

mRNA triplets	CGC	AAC
tRNA triplets		

Table 2.1

[1]

(c) Calculate the minimum number of DNA nucleotides required to code for a polypeptide with 238 amino acids. Show your working.

[2]

(d) Describe the role played by tRNA in polypeptide synthesis.

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.... [3]

Eukaryotic cells are able to regulate gene expression to ensure that resources within the cell are used effectively and efficiently.

(e) Regulation can be observed during the process of stem cell specialisation. Describe the most likely form of regulation taking place during stem cell specialisation, giving reasons for your answer.

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(f) Regulation can also occur following the synthesis of proteins. An example of such regulation is shown in Fig. 2.1 where inactive enzyme precursor pepsinogen is modified post-translationally to form active pepsin.

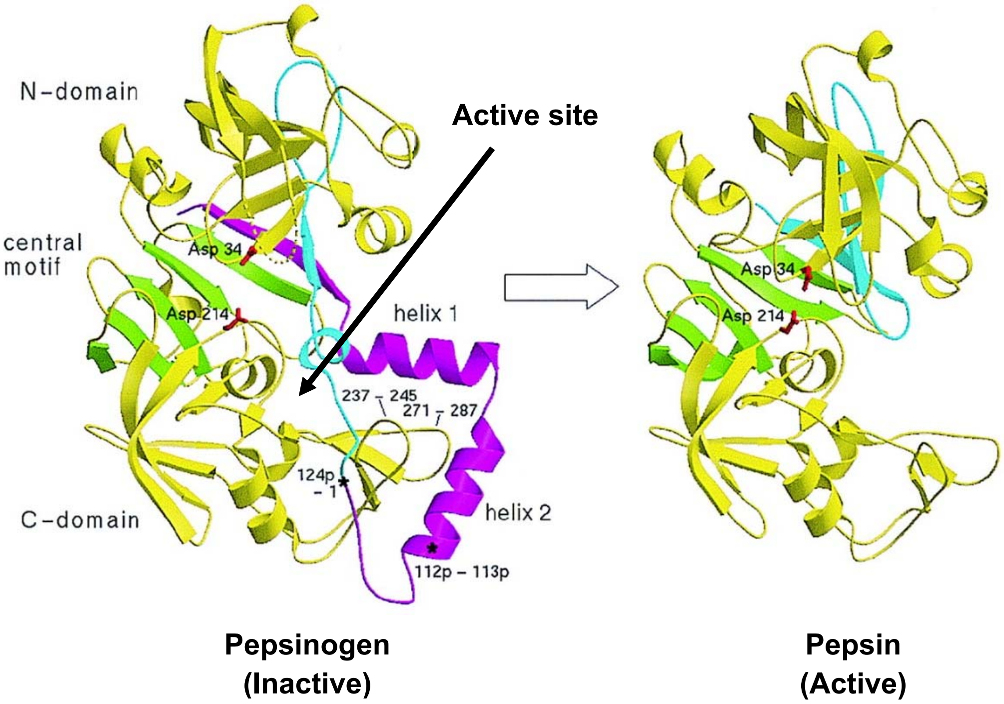

Fig. 2.1

(i) With reference to Fig. 2.1, identify the type of post-translational regulation observed in the enzyme and explain why it results in its activation.

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.... [2]

(ii) Suggest why pepsin is synthesised as an inactive precursor pepsinogen.

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.... [2]

[Total: 17]

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3. Bacterial cells can contain one or more plasmids which carry genes that are beneficial to its survival. One such plasmid contains *arg*, *his*, *leu* and *cys* genes that code for the biosynthesis of four essential amino acids; arginine, histidine, leucine and cysteine respectively.

A bacterial strain, strain **A**, with the following plasmid genes (*arg*+, *his*+, *leu*+ and *cys-)* was grown in nutrient media for several generations.

(a) State the amino acid(s), if any, that must be added to the nutrient media for strain **A** to grow.

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.... [1]

Strain **A** was then mixed with another bacterial strain, strain **B**, with the following plasmid genes (*arg*-, *his*-, *leu*- and *cys*+*)* and allowed to grow together at various time intervals before strain **B** was isolated and grown on nutrient media containing a variety of amino acids.

Table 3.1 shows whether colonies of strain **B** were observed on the various media at different time intervals (+ indicates the presence of an amino acid in the medium while - indicates its absence).

Time of incubation/ minutes	Supplementation of amino acids in medium				Presence of colonies
	Arg	His	Leu	Cys	
10	–	+	+	-	No
	+	-	+	-	Yes
	-	-	+	-	No
20	+	-	-	-	Yes
	-	-	-	-	No
25	-	-	-	-	Yes

Table 3.1

(b) Describe the process that has occurred between strains **A** and **B** when they were incubated together in the nutrient media.

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.... [4]

(c) Using information from Table 3.1, indicate the order of genes (*arg*+, *his*+, *leu*+) found on the plasmid in strain **A**. You should also indicate in your diagram the origin and direction of transfer.

[3]

(d) Using information from Table 3.1, explain your answer in **(c).**

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.... [3]

[Total: 11]

4. The fruit fly, Drosophila melanogaster, has eyes, a striped abdomen and wings longer than its abdomen. This is called a ‘wild-type’ fly.

Mutation has resulted in many variations of these features. Table 4.1 shows diagrams of a wild-type fly and three other flies, each of which shows one recessive mutation.

	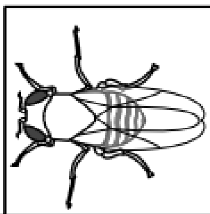	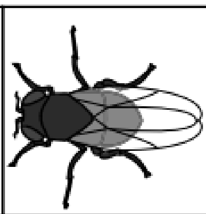	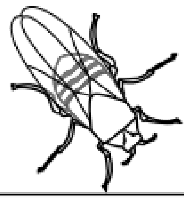	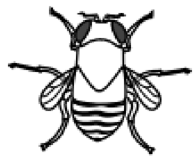
eyes	present	present	absent	present
abdomen	striped	black	striped	striped
wing description	long	long	long	short

Table 4.1

(a) Analysis of the mature mRNA formed from the mutant allele for black abdomen showed that one exon was missing although the length of both the mutant and normal alleles for abdomen colour were identical on the chromosome.

Suggest the genetic basis of this mutation and how it leads to a missing exon.

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.... [3]

(b) Using appropriate symbols, illustrate a cross between a fly without eyes and long wings and one with eyes and short wings that would result in **four** different phenotypes being observed in the offspring.

[5]

(c) A cross was carried out between a fly heterozygous for striped abdomen and long wings and a fly with a black abdomen and short wings. The results are shown below in Table 4.2.

offspring	number
striped abdomen long wing	86
black abdomen long wing	87
striped abdomen short wing	81
black abdomen short wing	78
total	332

Table 4.2

A chi-squared test (χ2) was carried out on these data. Complete Table 4.3 and calculate the value of χ2.

Observed Number (O)	Expected Number (E)	(O - E)	$(O - E)^2$	$(O - E)^2 / E$
86				
87				
81				
78				

Table 4.3

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

χ2: .... [3]

(d) Table 4.4 shows χ2 values.

degrees of freedom	probability						
	0.50	0.20	0.10	0.05	0.02	0.01	0.001
3	2.37	4.64	6.25	7.82	9.84	11.34	16.27

Table 4.4

Using Table 4.4, explain what conclusions can be made about the results of the χ2 test.

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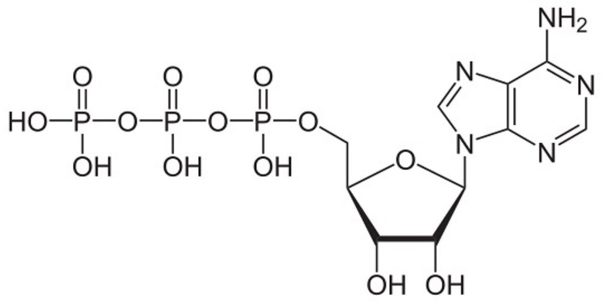
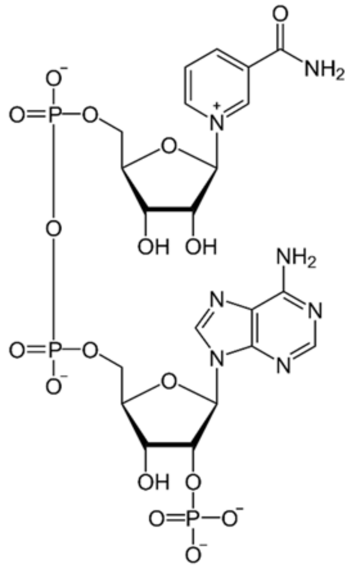
[Total: 13]

5. ATP and NADP both play important roles in photosynthesis. Fig. 5.1 represents the molecular structures of ATP and NADP.



ATP



NADP

Fig. 5.1

(a) Using Fig. 5.1, compare the structures of ATP and NADP.

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(b) Outline the roles of NADP in a cell.

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(c) State the names of the processes in which ATP is synthesised during photosynthesis and respiration respectively.

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(d) ATP serves as a source of energy for several metabolic processes in both photosynthesis and respiration. State two processes in **respiration** that requires ATP as an energy source.

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(e) The first substrate used in respiration is glucose. In a situation of excess glucose, some of these glucose is stored as fats instead of carbohydrates. Explain why animals prefer to store lipid instead of carbohydrates.

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[Total: 12]

6. Lactose intolerance is a condition in which individuals are unable to digest lactose as the cells in their small intestine are unable to synthesise sufficient quantities of the enzyme lactase. The undigested lactose will then interact with bacteria normally present in the large intestine and cause uncomfortable symptoms such as bloating, diarrhoea and gas production.

Lactose is commonly found in large quantities in dairy products. The distribution of adult population with lactose intolerance is shown in Fig. 6.1. A simplified world map illustrating major continents is shown in Fig. 6.2 for your reference.

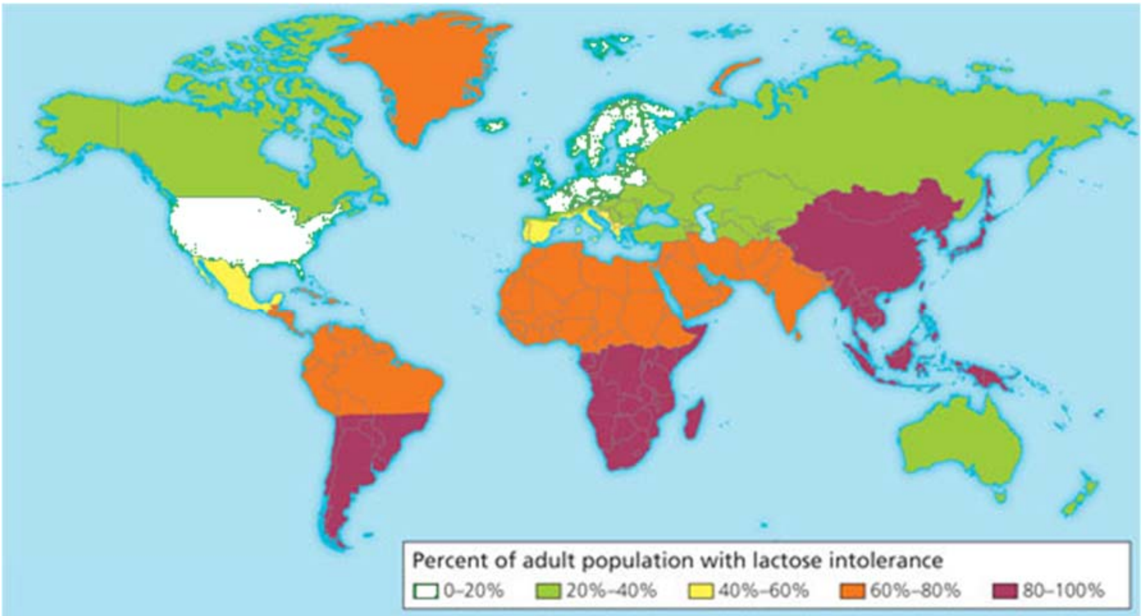

Fig. 6.1

Fig. 6.2

(a) With reference to Figs. 6.1 and 6.2, describe the distribution of lactose intolerance in adults.

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(b) Research has shown that most individuals are able to produce large quantities of lactase from birth, although this value can decline as the individual grows older due to environmental factors. The most prominent decline in lactase production is often observed when babies are switched from a predominantly milk-based diet to a solid food diet.

(i) Suggest why there is a prominent decline in lactase production in babies following a switch to a solid food diet.

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(ii) Using the information found in **(b),** suggest a reason for your observations made in **(a).**

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Level of expression of lactase is regulated by a lactase activator which stimulates expression of lactase in the presence of lactose. Human populations that inhabited the earth millions of years ago were found to express a lactase activator which is only weakly active. At present, most human populations express a more active version of this activator protein.

(c) Using your knowledge of evolution, explain the basis of this observation.

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[Total: 9]

7. Antigen presenting cells (APCs) such as macrophages are able to detect specific foreign antigens and present them to relevant adaptive immune cells. Recognition of these antigens require the use of specific receptors on the surface of these APCs.

(a) State the receptor found on macrophages that is used for antigen binding and recognition.

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Following binding, these antigens are then taken into the cell and processed for presentation to T cells.

(b) Describe how an **extracellular** antigen is subsequently presented to a T cell following binding to the receptor mentioned in **(a)**.

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Research has shown that a cell signalling pathway is triggered following binding of an antigen to its receptor. One of the more well understood signalling pathways involve a series of kinases, which are enzymes that catalyse the phosphorylation of its substrate. A common signalling pathway in APCs is shown in Fig. 7.1.

Recognition and binding of antigen to receptor on APC

↓

Binding and activation of MAP3 Kinase (MAP3K)

↓

Phosphorylation and activation of MAP2 Kinase (MAP2K)

↓

Phosphorylation and activation of MAP Kinase (p38) which is transported to the nucleus

Fig. 7.1

Activation of the signalling pathway shown in Fig. 7.1 leads to a large cellular response consisting of both cytokine production and also antigen presentation.

(c) Using your knowledge of signalling pathways and the information from Fig. 7.1, explain this observation.

[4]

[Total: 8]

8. Dengue and tuberculosis (TB) are two prominent infectious diseases of concern in many countries.

(a) Describe how TB is transmitted from an infected person to an uninfected person.

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Fig. 8.1 shows the distribution of dengue.

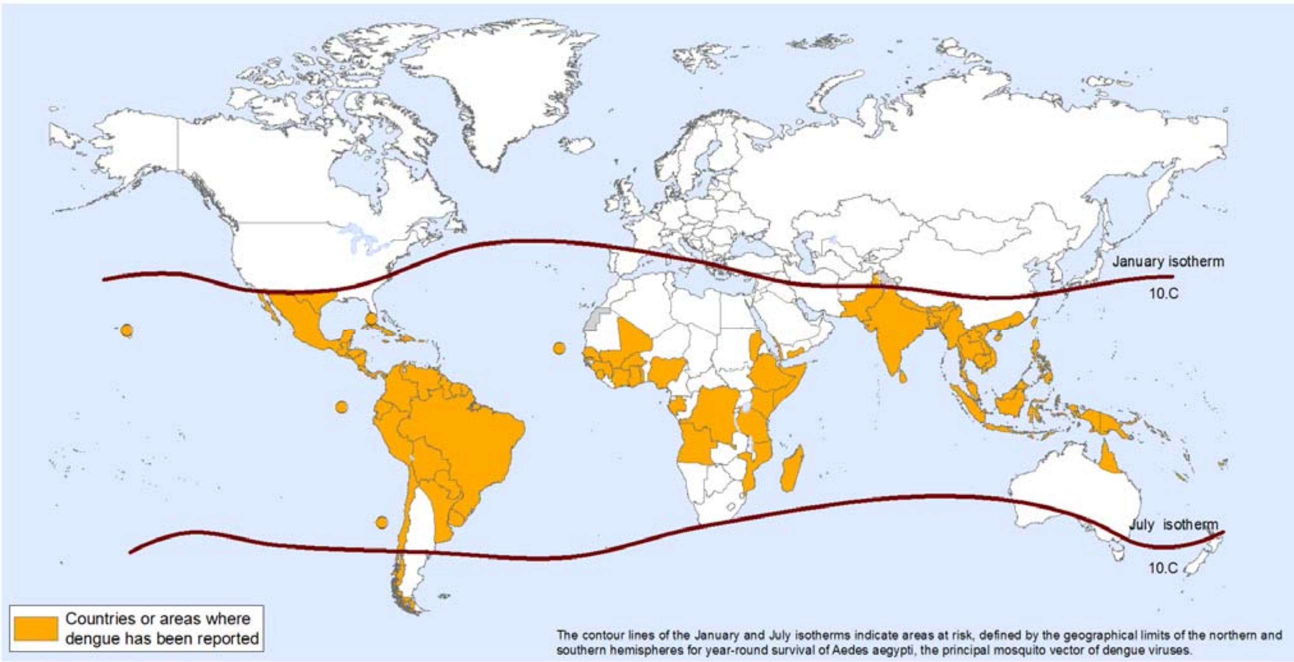

Fig. 8.1

(b) Unlike dengue, TB is found across the entire world. Explain why dengue shows the distribution shown in Fig. 8.1 whereas TB is found worldwide.

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Vaccinations are used to control infectious diseases. They were used as part of the programme to eradicate smallpox and as part of the continuing programmes against diseases such as polio and measles.

Smallpox was eradicated from the world in the 1970s. Polio is likely to be the next infectious disease to be eradicated.

(c) Explain how vaccination provides immunity as an important part of programmes to control and eradicate infectious diseases.

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Despite being a disease that has persisted for hundreds of years, there is currently no vaccine approved for the treatment of TB.

(d) Using your knowledge of the pathogenicity of *M. tuberculosis*, suggest why it is difficult to develop an effective vaccine for TB.

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Apart from the use of vaccination, other measures of controlling the spread of dengue involves the release of sterile male mosquitoes into areas with high dengue incidence. While this measure has successfully reduced mosquito populations dramatically, environmentalist are concerned about their potential detrimental ecological effects.

(e) Suggest one possible ecological concern that may arise from the use of sterile male mosquitoes.

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[Total: 12]

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